خدمتكم طريق خضناه لرضى الله

2021

ملخص
د.ريم دبابية

مختبر الكيمياء العامة



سنافر البوليتكنك | 0790640139

Exp (1) Lab Safety

By: Dr. Reem Dabaibeh.

- **Safety equipments**: أدوات السلامة .

(1) Fire extinguishers : used to extinguish small fires. Filled with CO_2, water, powder.

(2) Fire blankets : Used to extinguish small fires. It prevents the supply of oxygen to the fire.

(3) Safety shower: Used to extinguish the body in contact with fire or hazardous chemicals.

(4) Eye wash device: Used to wash eyes from any chemicals that might splash into it.

(5) First aid Kit: contains supplies used for medical treatment.

(6) Fume hood: Designed for experiments that produces hazardous or toxic vapors.

- **Personal Protective Equipments**: أدوات الحماية الشخصية

1) Safety googles
2) Lab coat
3) Gloves

- **Chemical bottles**: (stock bottles). عبوات الكيماويات

1. plastic bottles : (solid substances)
2. glass bottles : (liquid and solid)

- Never return unused chemicals back to stock bottles.
- All chemicals are treated as though it is very hazardous and dangerous.
- Learn the pictograms found on bottles of chemicals that show their hazardous effect.

(1) Flammable قابل للاشتعال
(2) Explosive ماده قابلة للانفجار
(3) Toxic مادة سامة .
(4) Corrosive ماده حارقة/كاوية

5. oxidizing : مادة مؤكسدة

6. Dangerous to the environment. خطيره على البيئة

7. Harmful / Irritant مؤذيه ، تسبب التحسس

8. Radioactive. مادة مشعة .

Balances: الموازين

1) Top-loading balance
accuracy = ∓ 0.01

2) Analytical balance ⇒ More accurate than top loading.
accuracy = ∓ 0.0001

- Never weigh very hot or very cold objects on the balance, it should be left to cool to room temperature.
- "Tare" button is to zero the balance. reading.
- Clean the balance pan after using it, and remove any spilled materials immediatly.

Bunsen Burner: هب بنسن

- Parts of Bunsen Burner:

1. Gas bulb "ساعة الغاز"
2. Gas hose "انبوب الغاز"
3. Gas valve "صمّام الغاز"
4. Gas vent "الجزء الذي يتحكم بكمية الاوكسجين"
5. Flame itself. "تكون على شكل مخروطين, الخارجي هو الأكثر حرارة"

- The flame preffered for heating is the blue flame, it has higher temperature than the yellow or red flame.

Blue flame	Yellow flame /red
• Higher temperature	• lower temperature
• nonluminous flame	• luminous flame.
• oxidized flame:	• reduced flame.
• Complete combustion of fuel	• Incomplete combustion
	• Dirty: leaves a layer of carbon on objects.

Dr. Reem Dabaibeh

- The fuel used is liquified potroleum which is a mixture of propane C_3H_8, and propane (C_4H_{10}).

- $C_4H_{10} + O_2 \longrightarrow CO_2 + H_2O$
  "complete combustion"

- $C_4H_{10} + O_2 \longrightarrow C_{(s)} + H_2O$
  " incomplete combustion"

Important Rules: تعليمات مهمة

- Consider all chemicals being hazardous.
- Toxic chemicals: methanol
- Flammable liquids: methanol
  ethanol
  chloroform
  benzene.
- Corrosive materials: Acids (HCl)
  Bases (NaOH)

- Flammable liquids should be heated in water bath, not on direct flame.
- Never pour water over acid, but acid over water.
- Use fume hood for carrying out experiments that produce hazardous vapors.
- Fill the pipette with pipette filler, not by mouth suction.

- **Glassware** : الأدوات الزجاجية
- equipments usually used to measure volumes

remember : $1\,L = 1000\,mL$

$1\,mL = 1\,cm^3$

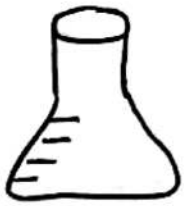
$1\,L = 1\,dm^3$.

(1) Beaker:
- approximate volume.
- Not used for measuring volumes.
- It is a reaction container or used for heating.



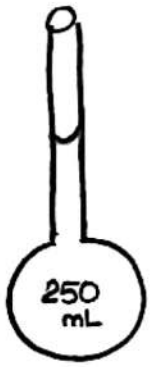
(2) Conical flask "Erlenmeyer flask"

- Approximate volume.
- Used for heating, and as rxn container.
- Used for titration experiments.



(3) Volumetric flask :

- Exact volume.
- Used for preparation of solutions.



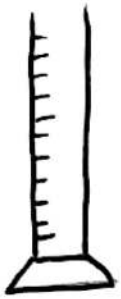
(4) Round - bottom flask :

- Not for volume measurments
- Used for heating liquids.



(5) Graduated cylinder:

- moderatly accurate for measuring volumes.

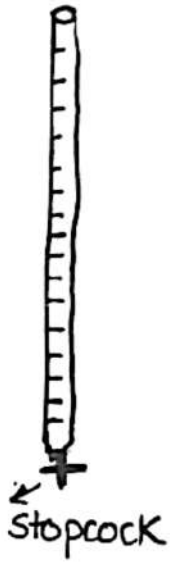


(6) pipette :

- Accurate for volume measurments
- more accurate than a cylinder.

(7) Burette:

- Accurate for volume measurement.
- Graduated from zero at the top to "25mL" or "50mL" at the bottom.
- One of the most accurate glassware.
- Used for titration experiments.



(8) Desiccator: to protect chemicals from humidity

(9) Test tube: to hold or mix samples

(10) Funnel: to funnel liquids from one container to another.

(11) Glass rod: used for stirring components of a reaction.

Dr. Reem Dabaibeh

Experiment (2): Physical Properties.
(Solubility and density)

(By. Dr. Reem Dabaibeh)

- Physical Properties: are properties for which there is no change in chemical composition of the material. (no change in material identity).

  Examples: density, melting point
  boiling point, solubility, color
  mass, hardness, ... etc.

- Chemical Properties: properties related to the chemical composition, and chemical reactivity of the material.

  Examples: oxidation reactivity
  chemical stability
  chemical reactivity
  Toxicity
  Flammability

(1) Solubility الذائبيـــة

- Solute : مذاب
- Solvent : مذيب
- polarity : قطبية المركب → polar
  → nonpolar.
- "Likes dissolve Likes" قاعدة الذائبيـــة

Solute	Solvent	
• polar	polar	⇒ Soluble
• nonpolar	nonpolar	⇒ Soluble
• polar	nonpolar	⇒ insoluble.

- Highly electronegative elements:
  F, O, N, Cl, Br, I عناصرها كهروسالبية عالية.
- less electronegative elements.
  H, C, Na, Mg, Al عناصرها كهروسالبية قليلة.
  (1A) (2A) (3A)
- In general in the periodic table Electronegativity increases from left to right "through a period"

- Electronegativity

Electronegativity decreases. ↓

→ Electronegativity increases.

(N O F)

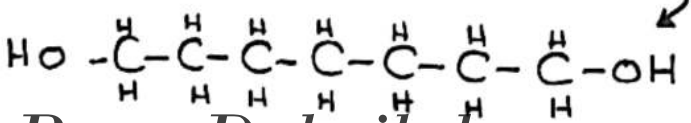
- Organic compounds :
  - C, H only ⇒ nonpolar.
  - C, H with "OH" or "NH_2" groups.

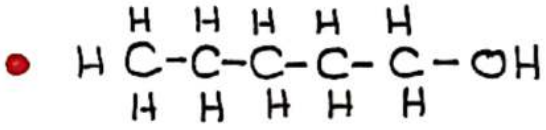
↙ polar if each "OH", "NH_2" group is connected to (5) carbons or less.

↘ nonpolar if each "OH" or "NH_2" group is connected to more than (5) carbons

Examples:

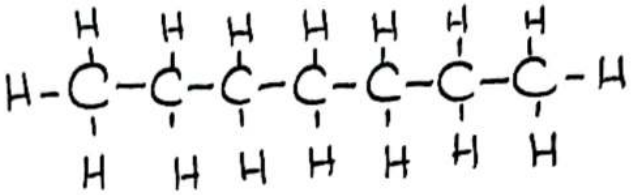
- CH_4 : nonpolar.
- CH_3OH: polar
- $CH_3CH_2NH_2$: polar
- $C_7H_{14}(OH)_2$ (polar)

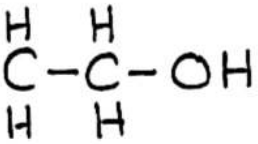


- $C_6H_{13}OH$ (nonpolar)
-  ↪ (polar)
- C_6H_6 (nonpolar)

Dr. Reem Dabaibeh

- Cyclohexanol  (nonpolar.)

- Hexane  (nonpolar)

- Ethanol  (C_2H_5OH) (polar)

- Ionic compounds:

  metal - nonmetal are usually polar.

  NaCl , KCl , $MgBr_2$, $CaCl_2$

- For a compound to be polar.

  1) Bond should be polar.

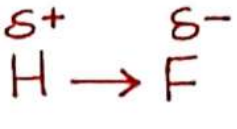
  " رابطة بين عنصر قليل لكهروسالبية له عالية وآخر له عنصر له كهروسالبية قليلة . "

  2) Net polar moment $\neq 0$.

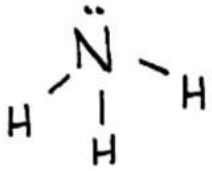
  شكل جزيء يكون له محصلة عزم قطبي $\neq$ صفر

Examples :

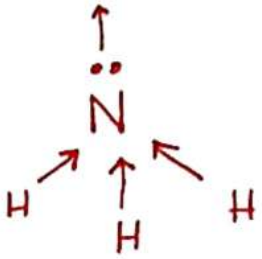
H – F (HF)



→ (polar)



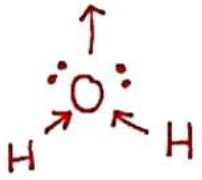
(NH_3)



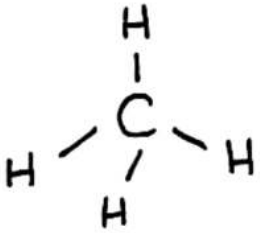
Bonds are polar.
(polar)



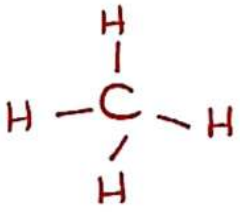
(H_2O)



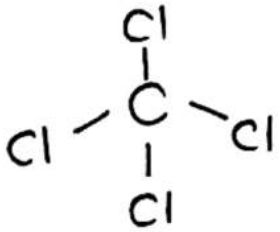
Bond is polar
(polar)



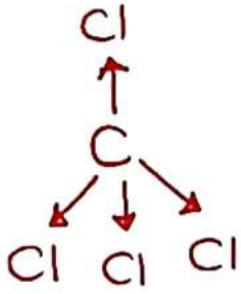
(CH_4)



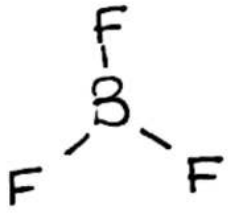
bonds are nonpolar.
(Nonpolar)



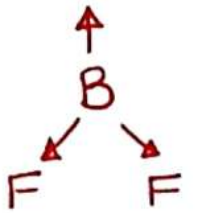
(CCl_4)



No polar momentum
(nonpolar)



(BF_3)



(nonpolar)

O = C = O (CO_2)



(nonpolar)

Dr. Reem Dabaibeh

- Molecules of identical atoms are nonpolar

  examples: I_2, O_2, O_3, N_2

- Acids and bases are usually polar.

  HCl KOH

  HCN $NaOH$

  H_2SO_4 $Mg(OH)_2$

  CH_3COOH (Bases)

  (Acids)

- Salts are usually soluble with some rules

IA: Na^+, Li^+, K^+, NH_4^+

7A: Cl^-, Br^-, I^-

NO_3^-

CH_3COO^-

SO_4^{2-} } Soluble.

↓

Except with

Ag^+

Pb^{2+}

Hg_2^{2+}

↓

Except with

Ag^+

Pb^{2+}

Ca^{2+}

OH^- CO_3^{2-}

S^{2-} PO_4^{3-}

} Insoluble

↓

Except with

IA , NH_4^+

Ca^{2+} , Ba^{2+} , Sr^{2+}

- Some examples:

Soluble	Insoluble
$NaNO_3$	$CaSO_4$
$(NH_4)_2SO_4$	$AgCl$
$CaCl_2$	$AgBr$
NaI	PbI_2
$AgNO_3$	Ag_2SO_4
$MgBr_2$	$Fe(OH)_3$
$Pb(CH_3COO)_2$	FeS
	$CaCO_3$
	$CaPO_4$

Solute / Solvent	NaCl	I_2	glucose.
Hexane			
cyclohexanol			
ethanol			
H_2O			

- Solubility depends on temperature

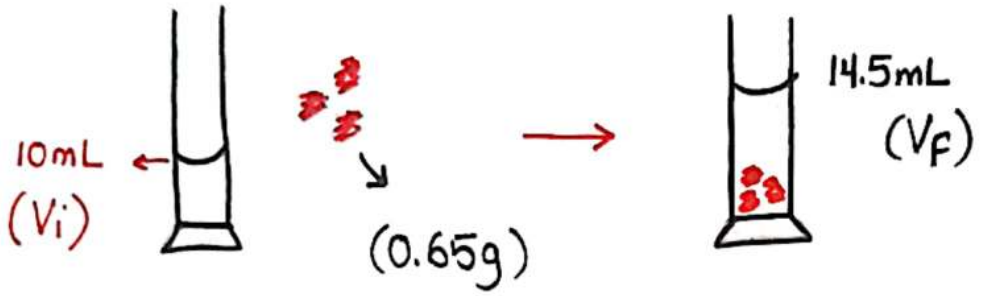
  As temp increases, solubility usually increases.

- Physical observation for solubility.
  - Colorchange.
  - Turbidity of solution.
  - precipitation of solute.
  - Immiscibility (sometimes two layers are formed).

Density of materials: "depends on temperature

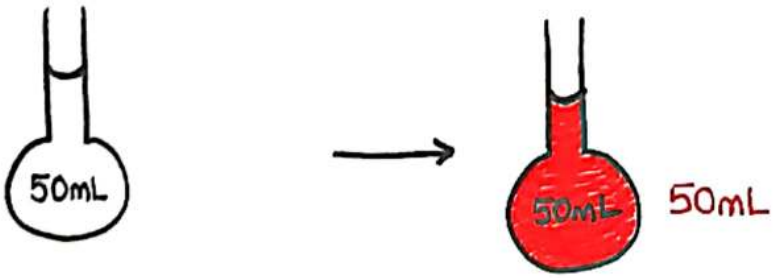
$$\text{Density} = \frac{\text{mass}}{\text{Volume}} \Rightarrow g/mL = g/cm^3$$

- For irregular solids:



$$\text{density} = \frac{\text{mass}}{\text{volume}} = \frac{0.65\,g}{4.5\,mL} =$$

- Finding density for a liquid:



$$d = \frac{22.22\,g}{50mL}$$

Empty volumetric flask

33.45g

Flask filled to the mark with the liquid

55.67g

$= 0.444\ g/mL$

Experiment (3) Physical Properties

- Boiling point: The temperature at which the substance is converted from the liquid state to the gaseous state.

- The temperature at which the vapor pressure of the liquid equals the external atmospheric pressure. $P_{vap} = P_{ext}$

- Normal boiling point: the boiling point of substance at standard conditions (1atm pressure)

- Boiling point for substance is measured for two purposes:

  (1) Identification of substance.

  (2) Determine the purity of a substance.

  " pure materials have sharp boiling points)

Factors affecting boiling point:

(1) The external pressure.

"As the external pressure increases, the boiling point increases"

(2) Molecular weight of the substance

"As the molar mass of compound increases, the boiling point will be higher.

(3) Intermolecular interactions between molecules of the substance.

As the intermolecular interactions between molecules is stronger, the boiling point will be higher.

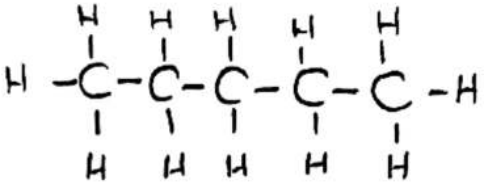
a. Hydrogen bonds.: (OH), (N-H), (F-H)

b. Dipole - dipole interactions : (polar compounds)

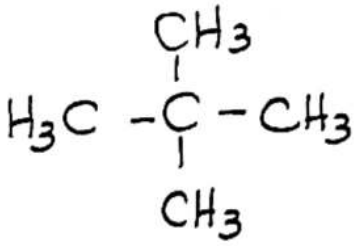
c. London forces : (between nonpolar compounds)

4. Structure of compound.

Branching of compounds decrease the surface area, the attractive forces become weaker, the boiling point decreases



(Higher boiling point)



(lower boiling point)

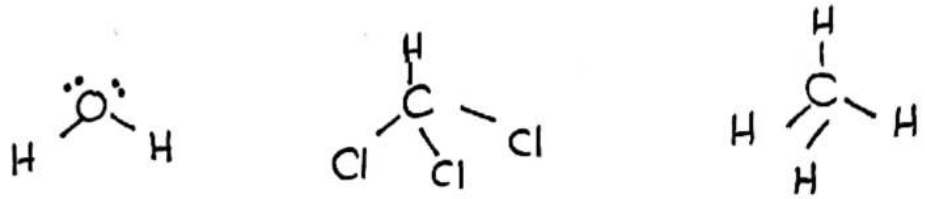
5. Presence of impurities.

Impurities

- Insoluble : "no effect on B.pt" (wood, glass, charcoal)
- Soluble impurities : "weakens the interactions, so boiling point decreases."
- Volatile impurities "decreases the boiling point"

- Example: Which has the highest B.pt?

H_2O or $CHCl_3$ or CH_4 ?



(polar) (H-bonds)

(polar)

(nonpolar)

$H_2O > CHCl_3 > CH_4$.

H - bonds
···O- H
F - H
···N - H

- Which has the highest boiling point?

CH_4 C_2H_6 C_4H_{10}

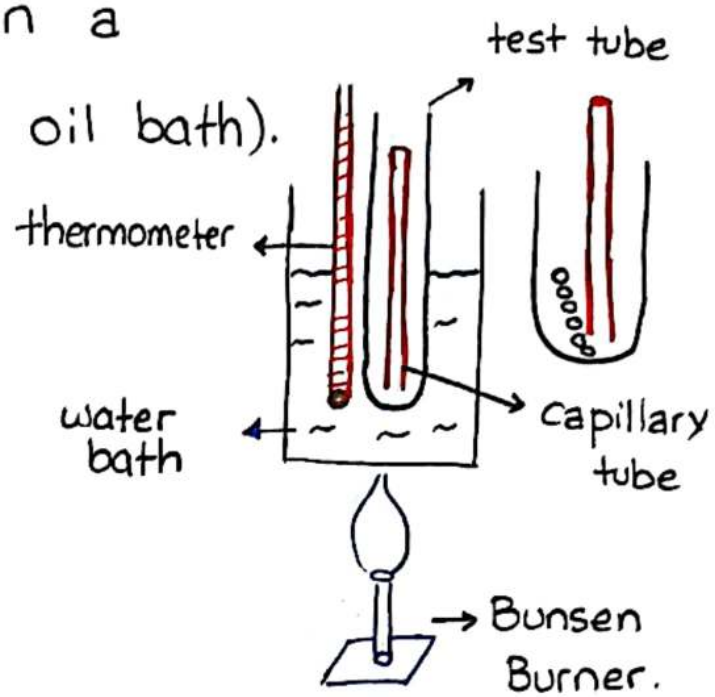
(16g/mol) (30g/mol) (58 g/mol)

- C_4H_{10} has the highest B.pt

All are nonpolar, but C_4H_{10} has the highest molecular mass.

Measurement of the boiling point.

- The liquid is heated in a water bath (or parafin oil bath).



- The liquid and thermometer should be at the same level.

- Boiling occurs when :

$P_{ext} = P_{vap.} \Rightarrow$ Temp = Boling point.

- This experimentally happen when you observe a stream of quickly moving bubbles coming out of the capillary tube.

- Use a water bath when B.pt $\leq 80°c$.
- Use a parafin oil bath when B.pt $> 80°c$

- Sometimes you observe that the liquid rised up in the capillary tube at the boiling point.

Evaporation and Boiling:

- Evaporation:
  - liquid gets converted to vapor.
  - It happens at the surface.
  - slow process.
  - spontaneous process
  - It could occur at any temp.
  - No bubbles observed.

- Boiling:
  - Liquid is heated continously.
  - Occurs over the entire mass of liquid.
  - Fast process. (quicker)
  - It happen a particular temp when $P_{vap} = P_{ext}$.
  - You can see bubbles.

- What is the effect of the following on the measured boiling point ?

(1) Rapid heating: fast heating will rise the temperature quickly, and the detected temperature will be more than the actual B.pt.

(2) Large quantity of liquid :

No effect.

(3) Impurities
- Soluble : decrease boiling pt.
- Insoluble: no effect.

(4) If thermometer and the test tube are not at the same level.

The liquid will feel a temperature different from the one the thermometer measures.

Experiment "4" : Water of Crystallization

"Formula of a hydrated salt."

- Hydrated Salt (hydrate) : Crystalline salts in which molecules of water are weakly attached to a salt.
  - General formula : $MA . x H_2O$ . . الصيغة العامة
    X: integer.
  - examples: $CuSO_4 . 5H_2O$.
    $BaCl_2 . 2H_2O$
    $Na_2CO_3 . 10 H_2O$.
    $Be(NO_3)_2 . 4 H_2O$

- Alums: specific type of hydrates.
  Alums are hydrated double sulfate salts.
  - General formula of alums:
    $M^+ M^{3+} (SO_4)_2 . X H_2O$
    $K Al (SO_4)_2 . 12 H_2O$
    $NH_4 Fe (SO_4)_2 . 12 H_2O$

M^+: monovalent ion
M^{3+}: trivalent ion.

Objective of the experiment: هدف التجربة:

1) Determine the mass percent of water in a hydrate

2) Determining the formula of an unknown hydrated salt "X". (moles of water).

- hydrate ≡ (hydrated salt) ≡ $Be(NO_3)_2.4H_2O$
- anhyrous: $Be(NO_3)_2$. ⇒ (بدون ماء)
- To find the value of "X":

$$\text{hydrated salt} \xrightarrow[\text{(heated)}]{\Delta} \text{anhydrous salt.}$$

- mass (hydrate) = (mass of anhydrous) + (mass of water.)
- Physical observations indicating that water has been removed.
  1. Color change تغيير اللون
  2. Froth تكوّن رغوة
  3. sample expansion. تمدد العينة

Dr. Reem Dabaibeh

During the experiment:

(1) The empty crucible should be dried well before being used and weighed.

(2) The hydrate is placed inside the crucible and heated to be dried (dehydration)

(3) The Bunsen flame should be blue colored to avoid formation of the black soot on the crucible

(4) After the sample is dried, the anhydrous sample should be cooled to room temperature before being weighed.

It is placed inside a dessicator, to prevent the sample to gain water molecules again from humidity of the surrounding.

- ضروري نترك العينة تبرد حتى توزنيها على الميزان، نتركها تبرد في المجفف في حو خاص عشان الرطوبة ما ترجع تعيد الملح جزيئات الماء مرة أخرى.

Dr. Reem Dabaibeh

- How to calculate "X" حسابات التجربة

$$X = \frac{\text{moles of water}}{\text{moles of hydrate}} = \frac{\text{moles of } H_2O}{\text{moles of anhydrous}}$$

- $\text{moles} = \frac{\text{mass}}{\text{molar mass}}$.

- $\%\ H_2O = \frac{\text{mass of } H_2O\ (g)}{\text{mass of hydrate}}$.

- Example:

$$CaSO_4 . 2H_2O \xrightarrow{\Delta} CaSO_4 + H_2O.$$

172 g/mol 136 g/mol 18 g/mol.

- Suppose you have 15g of $CaSO_4 . 2H_2O$. what is mass of $CaSO_4$ and H_2O in this sample

$$\text{moles (hydrate)} = \frac{15\ g}{172\ g/mol} = 0.087 \text{ mole}.$$

- moles of anhydrous ($CaSO_4$) = moles of hydrate ($CaSO_4.2H_2O$)

  = 0.087 mole.

- mass $CaSO_4$ = 0.087 mole × 136 g/mol

  = 11.8 gram $CuSO_4$.

- mass of H_2O = 15g (hydrate) − 11.8 (anhydrous).

  = 3.2 gram H_2O.

- $$\%\ H_2O = \frac{\text{mass } (H_2O)}{\text{mass(hydrate)}} \times 100\ \%$$

  $$= \frac{3.2\ g}{15\ g} \times 100\ \% = 6.6\ \%$$

ex) Suppose you have a hydrated salt of the formula $MA.X\,H_2O$.

If Formula weight of hydrate = 140g/mol and the hydrate is 25% water.

Find "x"

- 25% means 25g H_2O in 100g hydrate
  - 25g H_2O ↓ نخوّلها → moles H_2O
  - 100g hydrate ↓ نخوّلها → moles hydrate

$$X = \frac{\text{moles } (H_2O)}{\text{moles (hydrate)}}$$

$$X = \frac{(25/18)}{(100/140)} = 1.94 \approx 2.$$

Questions:

(1) mass of hydrate ($MgSO_4 . x H_2O$) = 3.211 g

mass of anhydrous $MgSO_4$ = 1.57 g.

a. $\% H_2O = \frac{(3.211 - 1.57)}{3.211} \times 100\%$

وزن الماء

وزن الملح كله

b. moles H_2O removed = $\frac{\text{mass}}{18\ g/mol}$

$= \frac{1.64\ g}{18\ g/mol} = 0.09$. (moles of water)

$X = \frac{0.09\ mol\ H_2O.}{(1.57 / 120.3)} = 6.8 \approx 7$.

moles of anhydrous $MgSO_4$ = 0.013 mol

water of Crystallization (X)

$MgSO_4 . 7 H_2O$

- If the hydrated salt is not completely dehydrated ⇒ إذا لم يتم إزالة الماء بشكل كامل كيف سيؤثر على نتيجة التجربة ؟

The mass of anhydrous will be more than the actual value.

$$X = \frac{\text{moles } (H_2O)}{\text{moles (anhydrous)}}$$ ⇒ المقام بكون كمية الكبريت الحقيقية الحقيقية

The value of "X" will be less than the actual value.

Exp (5) Molecular weight of a volatile liquid (Dumas Method)

- Volatile liquids: liquids that are easily converted in to their vapour state.
- These liquids have high vapor pressure and low boiling points ($\leq 90°C$) due to their weak intermolecular interactions.
- examples: ethers, alcohols, alkanes
  acetone, benzene, methane.
  methanol., diethyl ether.
- Ideal gas Law: قانون الغاز المثالي

$$PV = nRT$$

P: pressure of gas in (atm)

V: volume of gas in (L)

n: no. of moles of gas = $\frac{\text{mass}}{\text{M.wt}}$

R: Universal gas constant = $0.0821 \frac{\text{atm.L}}{\text{mol.K}}$

T: Temperature in Kelvin.

- 1 atm = 760 torr = 760 mmHg = 101.3 KPa.
- 1L = 1 dm^3 = 1000mL = 1000 cm^3
- ___ K = ___ °C + 273

- $PV = nRT \Rightarrow PV = \frac{m}{M.wt} \cdot R \cdot T$

$$Mwt = \frac{m \cdot R \cdot T}{P \cdot V} \quad (g/mol)$$

- Avogadro's Law of gases:

"Equal volumes of different gases contain equal number of molecules at the same conditions of temperature and pressure."

moles (gas "A") = moles of (gas "B")

at same T, P, V.

- **Dumas Method:**

  Procedure used to determine molecular weights of volatile organic liquids at room temperature.

- Objective of this experiment:

  To find molecular weight and density for an unknown volatile liquid using "Dumas" method.

- لمعرفة الكتلة المولية (والكثافة) لسائل متطاير سيتم تحويله إلى غاز، وتطبيق قانون الغاز المثالي عليه، لمعرفة كتلته المولية.

$$M.wt = \frac{m.R.T}{PV}$$

$$M.wt = \frac{d.R.T}{P}$$

- Given that three different gases (N_2, O_2, NH_3) occupy a 50 mL volumetric flask.

  If mass of N_2 (28 g/mol) is 60 gram. what mass of O_2 and NH_3 will occupy the same volume, at same temperature and pressure.

"المفروض بما أن الغازات الثلاث تحتل نفس الحجم عند نفس درجة الحرارة والضغط، أن تكون عدد مولاتهم متساوية."

- N_2 = 28 g/mol
- O_2 = 32 g/mol
- NH_3 = 17 g/mol

- $moles_{(O_2)} = moles_{(N_2)} = moles_{(NH_3)}$

$$\frac{m_{(O_2)}}{32} = \frac{60}{28}_{(N_2)} = \frac{m}{17}_{(NH_3)}$$

mass (O_2) = 68.57 g.

mass (NH_3) 36.43 g

Dr. Reem Dabaibeh

$$M.wt = \frac{m.R.T}{PV}$$

1) Initial weight = 95.732g. = flask + foil + rubber

2) Add 20mL liquid to the 100mL flask close with aluminum foil and make a small hole " to allow pressure bl lance inside and outside the flask.

$$P_{internal\ (gas)} = P_{external} = 747\ mmHg.$$

③ Heat the volatile liquid in water bath. "because it could be flammable"



④ The liquid will be totally converted into vapor. When this happens measure temp. of water ≈ 98°C

⑤ Cool the flask, and the vapor will be condensed upon cooling.



Take the final weight = 96.112g.

(flask + aluminum + rubber + gas)

Dr. Reem Dabaibeh

⑥ Fill the 100mL flask with water. (to the rim).

Then add the water into a cylinder to get the exact volume of the conical flask. = 90mL

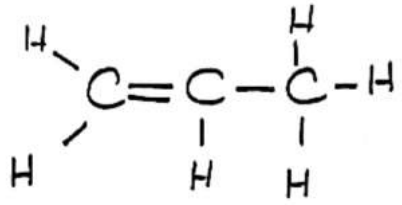
Ex) Two gases A, B are mixed in a 75mL flask at 755 mmHg and 30°C. If molar mass of "B" = 42g/mol Find the density of liquid "A" in (g/mL)

Experiment "6" : Empirical Formula

- Every compound has a "molecular formula" that shows the exact no. of atoms in this compound.

- The empirical formula of a compound is the formula that shows the simplest whole number ratio of atoms in the compound.

ex) propene:



$\Rightarrow$ C_3H_6 $\Rightarrow$ CH_2

Structural formula
صيغة بنائية

molecular formula
صيغة جزيئية

empirical formula.
صيغة أولية

- Ionic compounds have empirical formula identical to their actual formula. ex: Fe_2O_3 , $CaCl_2$

molecular formula	empirical formula	n
C_6H_6	CH	6
$C_6H_{12}O_6$	~~C_2H_2O~~ CH_2O ✓	6
H_3PO_4	H_3PO_4	1
Fe_2O_3	Fe_2O_3	1
C_4H_8	CH_2	4
C_3H_6	CH_2	3
C_5H_{10}	CH_2	5

- molecular formula = $(\text{empirical formula})_n$

  n: integer (عدد صحيح)

- $n = \dfrac{\text{molar mass of actual formula.}}{\text{molar mass of empirical formula}}$

- Different compounds could have identical empirical formulas.

"مركبات مختلفة قد يكون لها نفس الصيغة الأولية"

- How to find the empirical formula of a compound.

Example: A compound composed of 36.8% "N" and 63.2% oxygen.

If molecular wt of this compound = 76 g/mol. Find its molecular formul

لإيجاد الصيغة الأولية يجب تحويل الكتل إلى مولات

N	O	
36.8%	63.2%	⇒ If you suppose 100g Compound
36.8g	63.2g	
÷ 14 g/mol	÷ 16 g/mol	"نقسم الكتلة للعنصر على الكتلة الذرية"
2.62 mol	3.95 mol	
÷ 2.62	÷ 2.62	نقسم على أصغر عدد مولات
1	1.5	

$(NO_{1.5}) \times 2 = N_2O_3$

عند القسمة على أصغر عدد مولات ، لا نقرّب الأرقام دائماً .

- ـ.1 ـ.2 ـ.8 ـ.9

نقربها للأكبر نقرب للأصغر

- ـ.3 ـ.4 ـ.5 ـ.6 ـ.7

↓x3 ↓x2 ↓x2 ↓x2 ↓x3

- نضرب الصيغة كاملة بحدد صحيح للحصول على صيغة أولية فيها أرقام صحيحة .

For the example:

emp. formula = N_2O_3 (76g/mol)

من السؤال

$$n = \frac{76\ g/mol}{76\ g/mol} = 1$$

حسبها للصيغة الأولية

molecular formula = $(N_2O_3)_1 = N_2O_3$

- Given a 12.753 g sample of a compound composed of C, H, and "O". If the sample is combusted and produced 23 g of CO_2 and 7.857 g of H_2O. and given that its molecular wt = 226 g/mol.

Find its empirical and molecular formulas.

$$C_xH_yO_z \xrightarrow{O_2} CO_2 + H_2O$$

12.753 g → 23 g, 7.857 g

↓

mass of "C" +
mass of "H" +
mass of "O"

- لإيجاد الصيغة الأولية
يجب إيجاد عدد مولات كل من
الكربون والهيدروجين والاوكسجين

- مولات "C" في CO_2 = مولات "C" في العينة .
- مولات "H" في H_2O = مولات "H" في العينة
- لا يمكن حساب مولات "O" في العينة من H_2O أو CO_2

- 23g CO_2

↓

0.522 mol "CO_2"

↓ $CO_2 : C$ — 1 : 1 — 0.522 : x

0.522 mol "C"

↓

6.264 g "C"

- 7.857 g H_2O

↓

0.437 mol "H_2O"

↓ $H_2O : H$ — 1 : 2 — 0.437 : x

0.873 mol "H"

↓

0.873 g "H"

- To find moles of "O"

mass of "O" = 12.753g − (6.264g + 0.873g)

"وزن العينة" كاملة — (وزن الكربون) (وزن الهيدروجين)

mass "O" = 5.616 g.

moles "O" = $\frac{5.616\ g}{16\ g/mol}$ = 0.351 mol "O"

Dr. Reem Dabaibeh

C	H	O
0.522 mol	0.873 mol	0.351 mol
0.351	0.351	0.351
1.48	2.49	1
(1.5	2.5	1) x 2
3	5	2

$C_3H_5O_2$ (73 g/mol)

empirical formula (نسب كتلها المولية)

$$n = \frac{226.3\ g/mol}{73\ g/mol} = 3.1 \approx 3$$

(في السؤال)

molecular formula = $(C_3H_5O_2)_3$

= $C_9H_{15}O_6$

Dr. Reem Dabaibeh

Experimentally :

To find an empirical formula of a compound there are two methods :

(1) Synthetic method

ex. $Mg + O_2 \xrightarrow{\Delta}$ Mag. oxide.

(2) Decomposition method :

ex. $C_xH_zO_y \xrightarrow[O_2]{\Delta} CO_2 + H_2O$

- In this experiment, the objective is to find the emp. formula of an ionic compound (magnesium oxide)

- During the experiment :
  - Weigh the empty crucible.
  - add about 0.25 g magnesium turnings.
  - The magnesium is combusted using blue flame

- During combustion.
- Main rxn : $Mg + O_2 \longrightarrow Mg_xO_y$
  (gray colored)
- Side rxn : $3Mg + N_2 \longrightarrow Mg_3N_2$
  (white colored)
  (mag. nitride)
  (by-product)
- To remove the by-product
  few drops of water is added.

$$Mg_3N_2 + H_2O \longrightarrow Mg(OH)_2 + NH_3\uparrow$$

ammonia.

Δ ← إعادة تسخين حتى يتم التخلص من الماء الزائد.

$$(Mg_xO_y) + H_2O$$

(أكسيد المغنيسيوم)

- The crucible is left to cool before weighing the final weight.

Experiment (7) : "Reactions of Aqueous solutions"

Precipitation reactions (Limiting reactant)

- Solution: mixture of solute + solvent.
- Aqueous solution: solvent is water.
- Precipitation reaction.

$$CaCl_{2\,(aq)} + Na_2CO_{3\,(aq)} \rightarrow CaCO_{3\,(s)} + 2NaCl_{(aq)}$$

- Molarity of solution = $\frac{\text{moles (solute)}}{\text{Liters of solution}}$
- moles of solute = $M \times V_L$
- Limiting reactant: reactant that is consumed firstly " it is totally consumed"
- Limiting reactant: determines amount of products produced.

Dr. Reem Dabaibeh

For the rxn:

$$2P_{(s)} + 3Cl_2 \longrightarrow 2PCl_3$$

If 5.0g P (31g/mol) are allowed to react with 7.0 g Cl_2 (71 g/mol).

(a) Which reactant will determine amount of PCl_3 (137.5 g/mol) formed.

(b) What is mass of PCl_3 that is formed.

$$\frac{\text{moles } Cl_2}{3} = \frac{\text{moles } PCl_3}{2}$$

moles $PCl_3 = \frac{0.098 \times 2}{3}$

moles $PCl_3 = 0.065$ mol

mass PCl_3 =

$= 0.068 \times 137.5$ g/mol

$= 8.98$ g PCl_3.

P	Cl_2
$\frac{5g}{31}$	$\frac{7g}{71}$
↓	↓
0.161 mol	0.098 mol
↓ ÷2	↓ ÷3
0.081	0.033
	الناتج الأصغر هو عثل العامل المحدد

Suppose you have the following reaction:

$$3\,AgNO_{3\,(aq)} + AlCl_{3\,(aq)} \rightarrow 3\,AgCl_{(s)} + Al(NO_3)_{3\,(aq)}$$

If ~~50mL~~ 500mL (0.2M) $AgNO_3$ are allowed to react with 80mL (0.1M) of $AlCl_3$ solution.

During reaction 2.98 g of AgCl has been precipitated. (AgCl = 143.5 g/mol)

Calculate percentage yield of reaction

- The L.R is $AlCl_3$.

$$\frac{\text{moles } AlCl_3}{1} = \frac{\text{moles } AgCl}{3}$$

moles AgCl = moles $AlCl_3$ × 3

= 0.008 × 3

= 0.024 mol AgCl

× 143.5 g/mol → 3.44g AgCl

$AgNO_3$	$AlCl_3$
$\frac{500}{1000} \times 0.2$	$\frac{80}{1000} \times 0.1$
↓	↓
$\frac{0.1\text{ mol}}{3}$	$\frac{0.008\text{ mol}}{1}$
↓	↓
0.033	0.008
↓	↓
Excess.	L.R.

actual yield of AgCl = 2.98 g.

Theoritical yield of AgCl = 3.44 g.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoritical yield}} \times 100$$

$$= \frac{2.98\,g}{3.44\,g} \times 100$$

$$= 86.6\ \%$$

In this experiment:

you mix two solutions:

25mL (0.5M) $CaCl_2$ solution and

20mL (1.5 M) Na_2CO_3 solution.

$$CaCl_{2\,(aq)} + Na_2CO_{3\,(aq)} \longrightarrow \underset{\text{(white ppt)}}{CaCO_{3\,(s)}} + 2NaCl_{(aq)}$$

moles $CaCl_2$ = moles Na_2CO_3

moles $CaCl_2$ = $\frac{25}{1000}$ x 0.5M = 0.0125 mol

moles Na_2CO_3 = $\frac{20}{1000}$ L x 1.5 M = 0.03 mol

- $CaCl_2$ is L.R.

moles $CaCl_2$ = moles $CaCO_3$ = 0.0125 mol.

mass of $CaCO_3$ produced

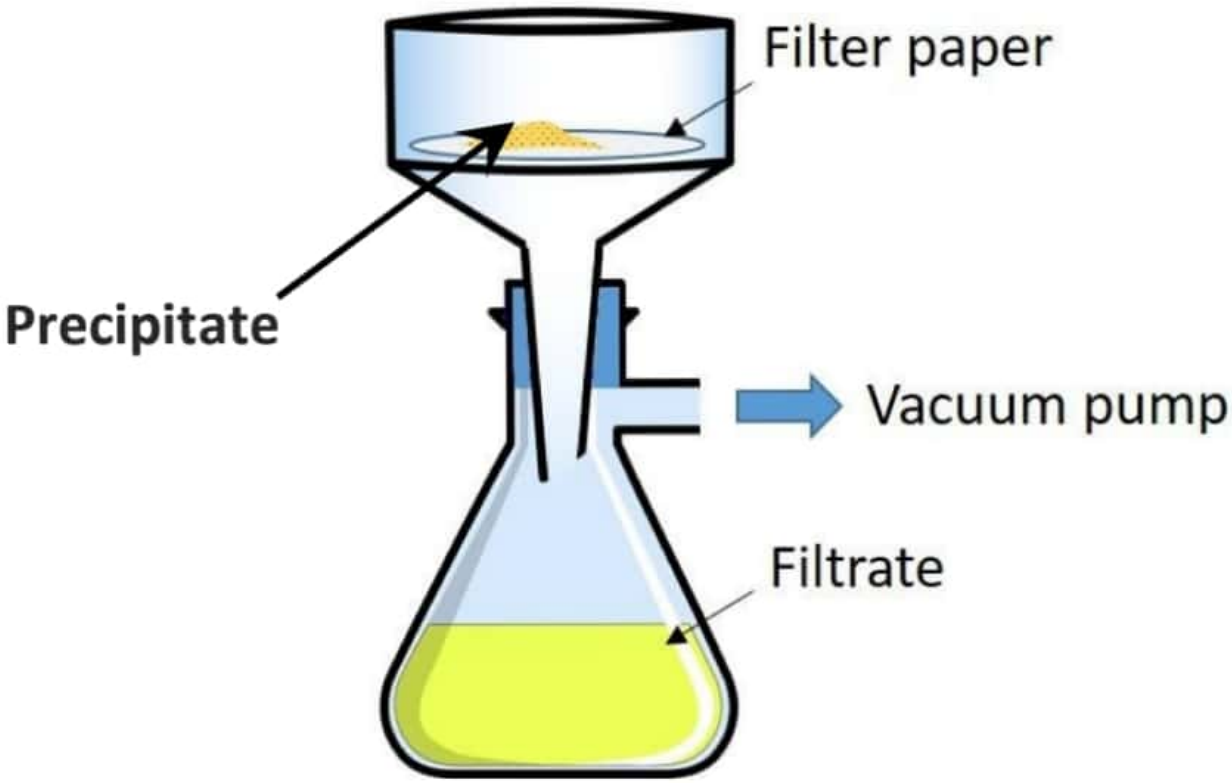
(theoritical yield) = 0.0125 mol x 100 g/mol

= 1.25 gram.

If 0.92 g is actually produced.

Percentage yield = $\frac{0.92\ g}{1.25\ g}$ x 100

= 73.6 %


Filter paper
Precipitate
Vacuum pump
Filtrate

Experiment "8" : Acid - Base Titration

"Vinegar Analysis"

- Titration: Technique used in labs to determine the concentration of a solution by reacting it with a standard solution.

- standard solution: a solution of a known concentration (exact concentration)

- For any titration we have two solutions:

  1) Titrant : known concentration "standard solution.

  2) Titrate : the solution of unknown concentration.

- There should be a balanced equation between these two solutions, to find out the unknown concentration.

Titrant + titrate → products.

- In titrations: we usually need a reagent called "indicator" which is added to the solution, to detect the end point of reaction.

- End point of titration: the point at which moles of titrant are equivalent to moles of titrate "according to the balanced chemical equation"

- At the end point, a physical observation could be detected as:
  - color change.
  - a gas produced.
  - a precipitate forms.

  All these indicate that the reaction is complete.

Instruments used for titrations:

(1) Burette ⇒ Filled with titrant
"standard solution"

(2) Conical flask (recieving flask)
filled of the solution of unknown concentration.

Types of titrations:

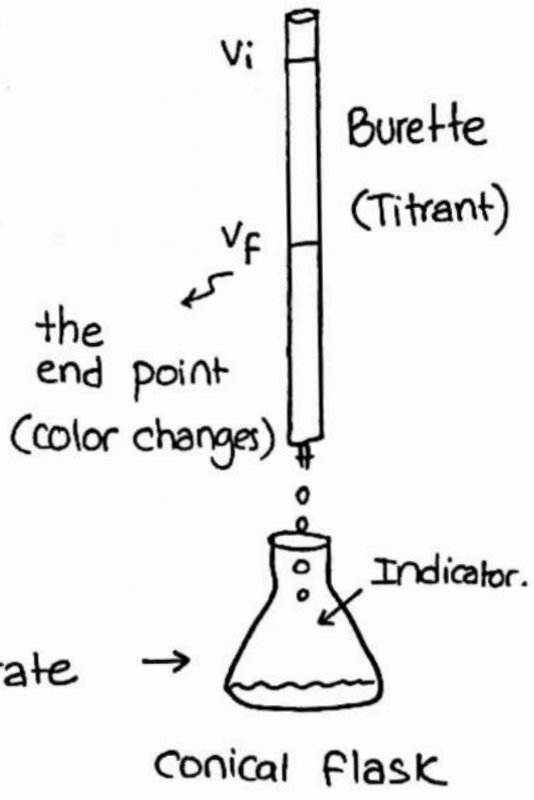
(1) Acid - Base titration (Neutralization titration)

(2) Redox titration.

(3) Precipitation titration.

(4) Complexometric titrations.

- Remember: the titrant should have an exact known concentration.



Vinegar Analysis:

- Vinegar: a diluted solution of acetic acid. (CH_3COOH)

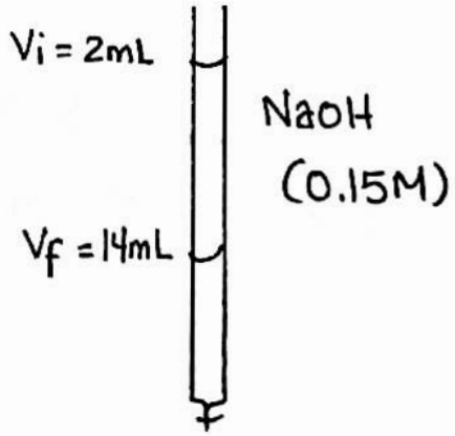
- To get the mass percent of CH_3COOH in vinegar, it is titrated with a base ($NaOH$).

- $NaOH + CH_3COOH \rightarrow CH_3COONa + H_2O$

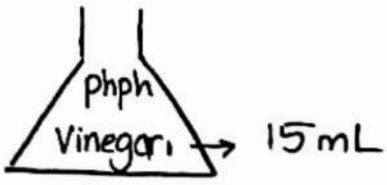
  moles ($NaOH$) = moles (CH_3COOH)

- Titrant: $NaOH$ (0.15M)
  "burette"

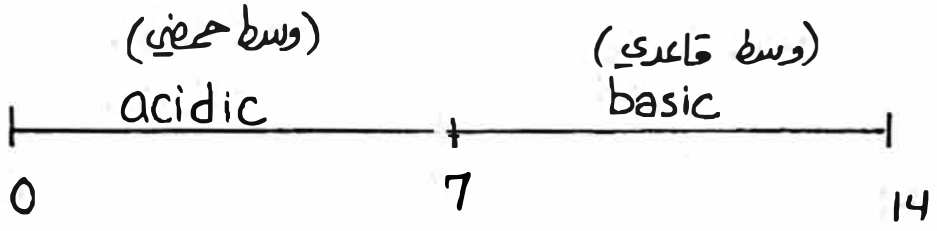
- Titrate: Vinegar solution
  (1 mL)
  Contains (CH_3COOH)



- Indicator: phenolphthalein
  (Phph)
  - Colorless (acidic medium)
  - pink (basic medium)



- The color of the indicator depends on the pH of solution.



phph ⇒ (colorless) متعادل (pink)

- At the end point of titration:
  - moles of NaoH = moles CH_3COOH (حسب المعادلة)
  - The color changes from colorless to pink
  - The reaction is complete "no CH_3COOH is left in the flask"

- For the previous titration.

M (NaoH) = 0.15M

V (NaoH) = $V_f - V_i$ = 14 - 2 = 12 mL.

V (Vinegar) = 15mL

moles (NaOH) = moles (CH_3COOH).

M x V_L = moles (CH_3COOH).

$0.15\,M \times \frac{12\ mL}{1000}$ =

0.0018 mol = moles CH_3COOH.

$$M_{CH_3COOH} = \frac{\text{moles } (CH_3COOH)}{V_L \text{ (vinegar)}}$$

$$= \frac{0.0018 \text{ mole}}{15\ mL \div 1000} = 0.12\ M.$$

$$\%\ CH_3COOH = \frac{\text{mass } (CH_3COOH)}{\text{mass (vinegar)}} \times 100.$$

$$= \frac{\text{moles} \times M.wt}{V_{(Vinegar)} \times \text{density}_{(Vinegar)}} \times 100\ \%$$

$$= \frac{0.0018 \times 60\ g/mol}{(15mL \times 1.04\ g/mL)} \times 100\ \%$$

= 0.7 %

Procedure:

(1) Rinse the burette with distilled water.

(2) Rinse it with small amount of titrant (NaOH)

(3) Fill it with NaOH. (Read Vi)

(4) Fill the flask with 15mL of vinegar. after washing it well.

(5) Add 5 drops of phph. (indicator)

(6) start titration with NaOH.
The color will stay colorless until it changes to pink.

(7) when the pink color appears, stop titration ⇒ you reached the end point.

(8) Read "V_f" of NaOH

(9) Make your calculations.

In case the analyzed vinegar sample is diluted with water.

5mL Vinegar Solution "original solution" "from bottle" V_1 $\xrightarrow[\text{water.}]{+\ 10mL}$ 15mL Vinegar Solution "diluted solution" "titrated" V_2

$$\text{dilution factor (dF)} = \frac{V_2}{V_1} = \frac{15}{5} = 3$$

$$M_{CH_3COOH\ (original)} = M_{CH_3COOH\ (diluted} \times 3$$

$$= 0.12 \times 3\ M$$

$$= 0.36\ M.$$

↘

(داخل العبوة) للعينة الأصلية

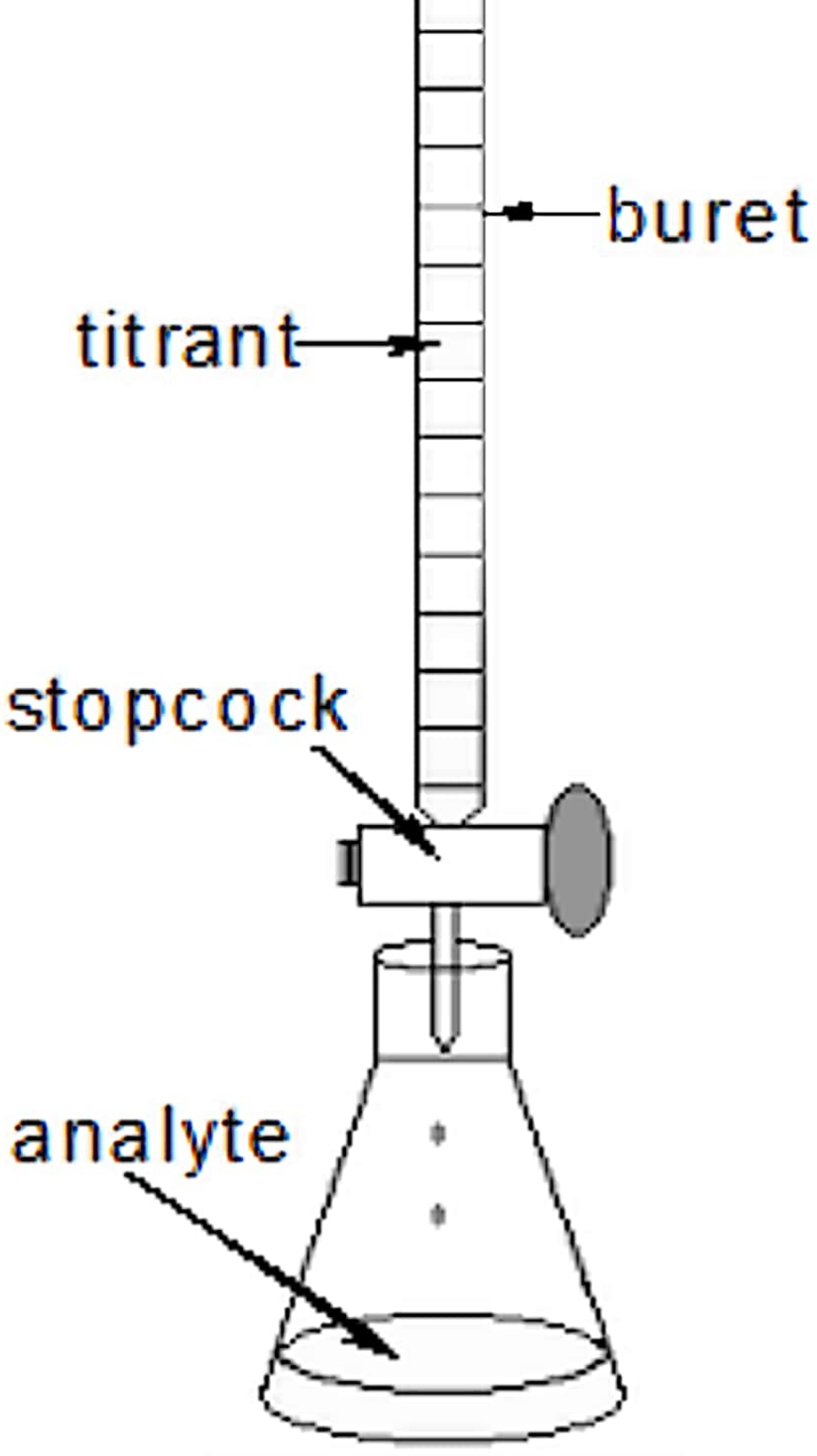
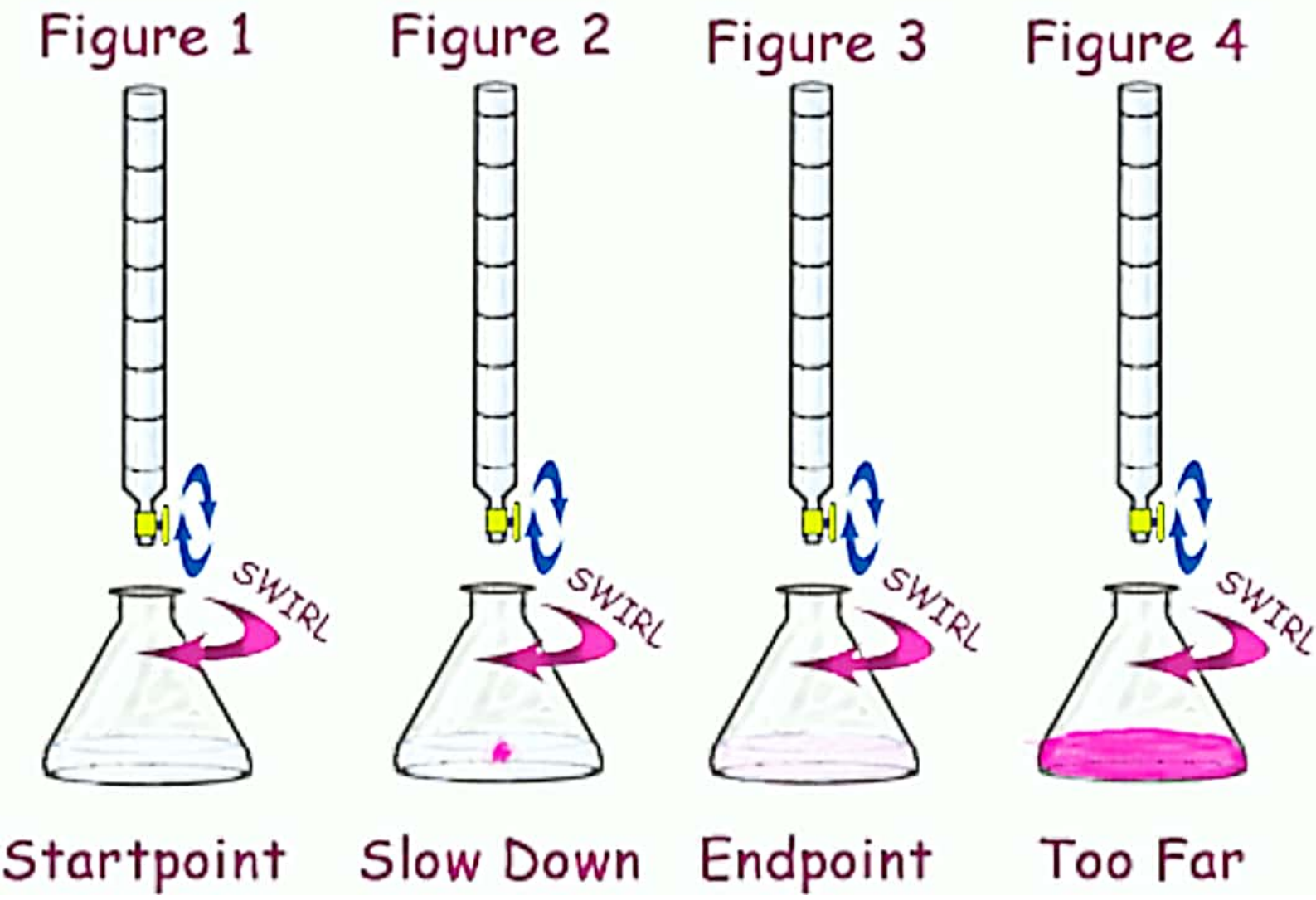


Figure 1: Titration Setup

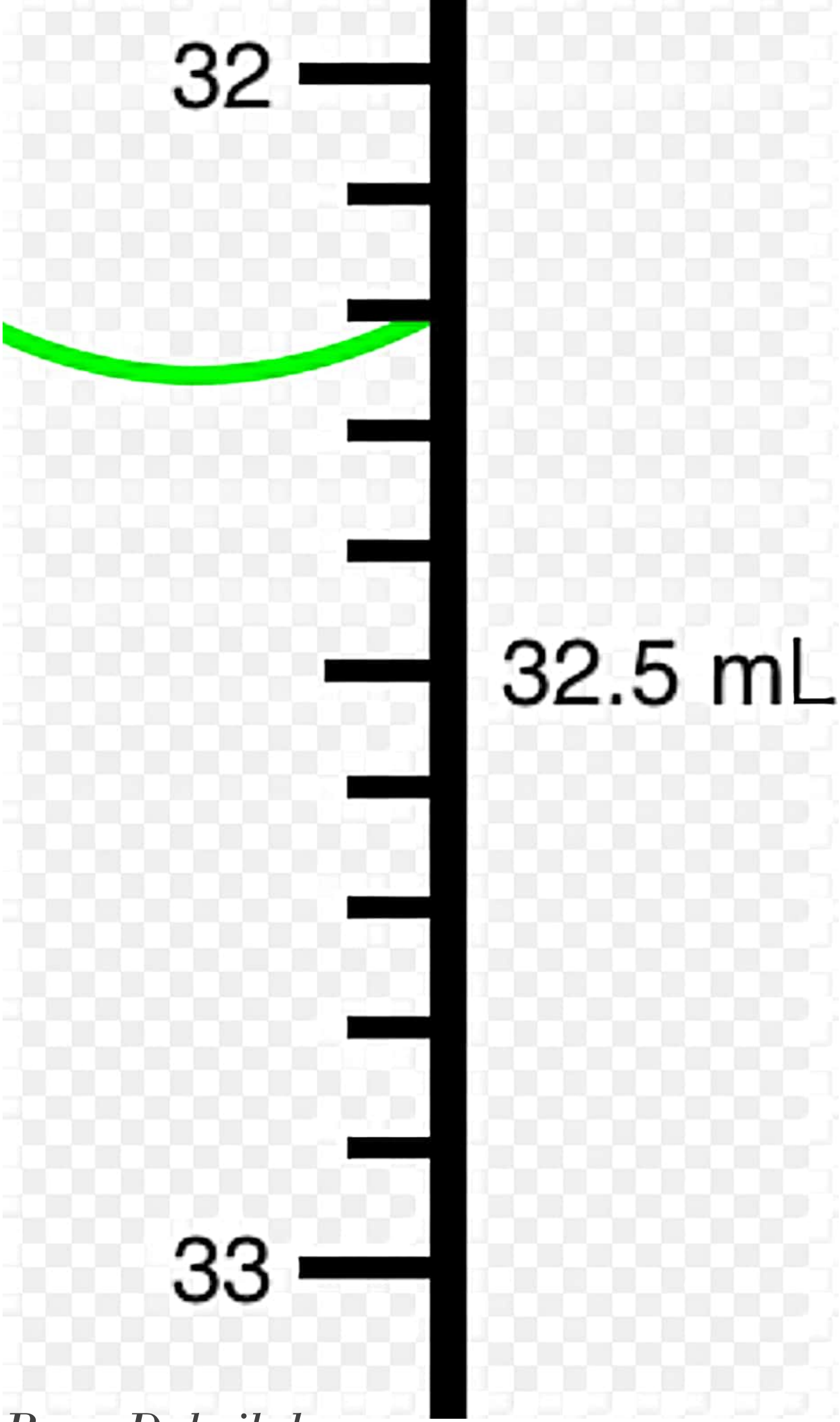
Dr. Reem Dabaibeh

Titration of an Acid with a Base

using phenolphthalein indicator



Dr. Reem Dabaibeh



Dr. Reem Dabaibeh

Experiment "9" : "Volumetric Analysis"

"Redox Titration"

"Bleach Analysis"

- The active ingredient in bleach is sodium hypochlorite ($NaOCl \equiv OCl^-$)
- To determine amount of (OCl^-) in a bleach sample, a reduction-oxidation reaction of OCl^- with I^- is performed.
- NaOCl (OCl^-) is the oxidizing agent.
- KI (I^-) is the reducing agent.
- (OCl^-) is reduced to (Cl^-)
- (I^-) is oxidized to (I_2)

The redox rxn:

$$OCl^-_{(aq)} + 2I^-_{(aq)} + 2\,H_3O^+_{(aq)} \longrightarrow I_{2(aq)} + Cl^-_{(aq)} + 3H_2O_{(l)}$$

عامل مؤكسد . عامل مختزل . catalyst (يسرّع التفاعل)

Dr. Reem Dabaibeh

$$OCl^- + 2I^- + 2H_3O^+ \longrightarrow I_2 + Cl^- + 3H_2O$$

- OCl^-: unknown conc. (bleach)
- H_3O^+: acid added to speed up rxn.
- I_2: reacts with the titrant.

- I_2 produced reacts with titrant.
- Titrant: Sodium thiosulfate.

  $Na_2S_2O_3 \equiv S_2O_3^{2-}$

  Known conc. = 0.1 M

  Filled in the burette.

- This is indirect redox titration

In general:

OCl^- (analyte) $\longrightarrow I_2$

$I_2 + S_2O_3^{2-} \longrightarrow I^-$

- I_2: brown color
- I^-: Colorless.

- المادة المراد تحليلها لاتتفاعل بشكل مباشر مع المادة المعايرة (titrant) بل يوجد مادة هي وسيط (I_2) هي التي تتفاعل مع المعاير ($S_2O_3^{2-}$)

In back titration: (two reactions)

(1) $OCl^- + 2I^- + 2H_3O^+ \longrightarrow Cl^- + I_2 + H_2O$

(2) $I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

I_2: oxidizing agent; $S_2O_3^{2-}$: reducing agent

From rxn (1)

$$\text{moles } (OCl^-) = \text{moles } (I_2)$$

From rxn (2)

$$\text{moles } (I_2) = \frac{\text{moles } S_2O_3^{2-}}{2}$$

$$= \frac{(M \times V_L)_{S_2O_3^{2-}}}{2}$$

So: $$\text{moles } (OCl^-) = \frac{\text{moles } S_2O_3^{2-}}{2}$$

Important notes:

(1) 5mL bleach (OCl^-) + 2g KI (I^-) + HCl

⇒ I_2 solution is produced.

(brown color)

(2) place I_2 in darkness to avoid photooxidation of (I^-)

It should be left for 5 minutes.

(3) Fill burette with 0.1M $S_2O_3^{2-}$ and read (V_i)

(4) Start titration until brown color becomes yellow.

Here most (I_2) is consumed.

but not all of it.

(5) Add starch (indicator)

A blue color should appear.

If not ⇒ means you have passed the end point.

(6) Continue titration until blue color disappears (end point).

- The blue color that appears is due to formation of an I_2- starch complex.
- If the blue color doesn't appear, that means you have passed the end point (repeat all your work)
- If bleach + I^- is not placed in a dark place, photooxidation will happen.
- The end point of titration is when

  blue color turns out to colorless

  ↓ ↓

  (I_2- starch) (I^-)

- The color change from brown to yellow means that "I_2" is reacting with "$S_2O_3^{2-}$", and I_2 is being consumed gradually.

Calculations:

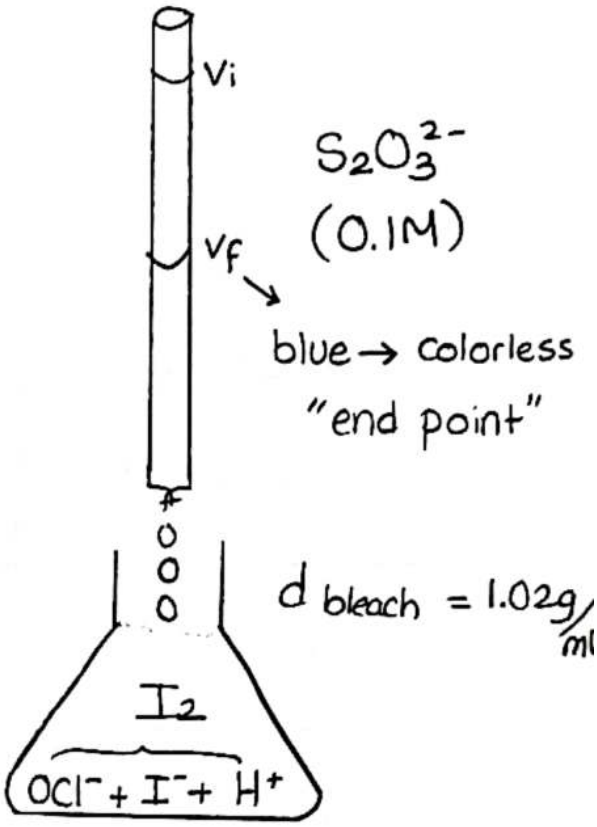
(1) moles $S_2O_3^{2-}$ = M x V_L.

(2) moles I_2 = $\frac{\text{moles } S_2O_3^{2-}}{2}$

(3) moles OCl^- = moles I_2

(4) diluted bleach = 25 mL

(5) dilution factor = 20.



(6) Concentrated bleach. = $\frac{25}{20}$ = 1.25 mL.

(7) Molarity of diluted bleach = $\frac{\text{moles } OCl^-}{25 \times 10^{-3} \text{ L.}}$

(8) $M_{\text{conc. bleach}} = M_{\text{diluted}} \times 20$

(9) $\%NaOCl = \frac{\text{moles } (OCl^-) \times 74.5 \text{ g/mol}}{1.25\text{mL} \times 1.02\text{g/mL}} \times 100.$

(1.25mL ← حجم المحلول الأصلي) (1.02g/mL → كثافة المحلول)

	Vinegar Analysis	Bleach Analysis
Sample:	Vinegar	Bleach.
Active ingredient (analyte)	CH_3COOH (acid)	NaOCl (oxidizing agent)
Titrant	NaOH	$Na_2S_2O_3$
Indicator	phenolphthalein	Starch.
End point	colorless → pink	blue → colorless
Type of Titration	Acid - Base titration.	Redox titration
Titration	Direct titration	Back titration.
Stoichiometery	$CH_3COOH : NaOH$ 1 : 1	$OCl^- : S_2O_3^{2-}$ 1 : 2

Dr. Reem Dabaibeh

- $2\,OCl^- \longrightarrow Cl_2$.

$$\%\ Cl_2 = \frac{\left(\frac{\text{moles } OCl^-}{2}\right) \times 71\,g/mol}{1.25mL \times 1.02\,g/mL} \times 100.$$

Note: 5mL is taken from original bleach. diluted to 100mL.

$$5mL \iff 100mL$$
$$x \iff 25\,mL$$

$x = 1.25\,mL$ (original bleach).

- dilution factor $= \frac{V_{dil}}{V_{conc}}$

$$= \frac{25}{1.25}$$

$$= 20.$$

Experiment "10" Thermochemistry and Calorimetery.

- Objective: to measure enthalpy of a rxn.
- Thermochemistry: branch of chemistry that measures heat gained or evolved during chemical process.
- Calorimetery: an experimental method used to measure enthalpy of a reaction by measuring temperature change of surrounding during the rxn.
- Enthalpy: heat content of reaction at constant pressure.
- ΔH_{rxn}
  - (+ve) : rxn is endothermic
  - (-ve) : rxn is exothermic.
- In calorimetery (direct way to measure ΔH The surrounding should be specified.

Calorimetery :

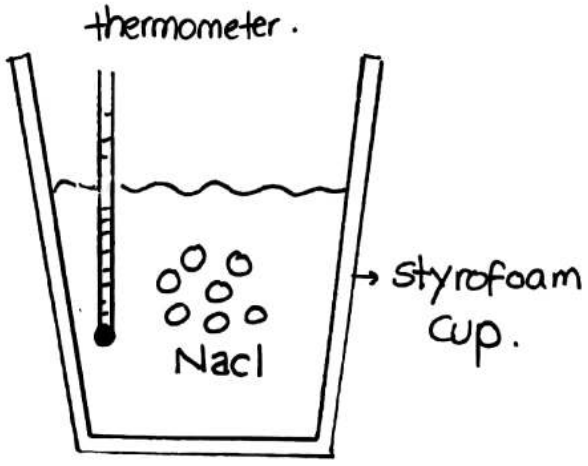
(1) open system: Cup-Calorimeter.

- pressure is constant.
- Used for solution reactions.
- The cup used is usually from styrofoam.
- The surrounding is :

$$q_{surr} = q_{air} + q_{cup} + q_{new\ solution}$$

$$NaOH_{(aq)} + HCl_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(\ell)}$$

$$q_{new\ soln} = m \times sp.ht \times \Delta t$$



- m: mass of solution

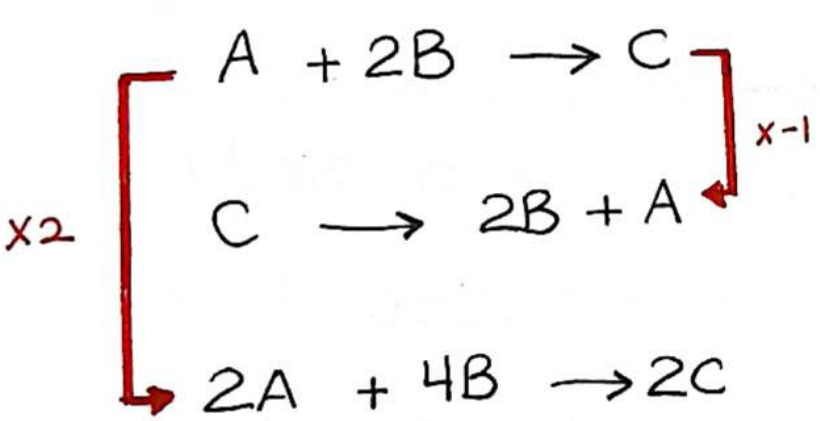
$$= V_{(mL)} \times d_{(g/mL)}$$

- Sp.ht: specific heat of new solution
- $\Delta t = t_f - t_i = t_{max} - t_i$

$$q_{p\,(rxn)} = -q_{(surr)}$$

$$\Delta H_{rxn} = \frac{q_{p\,(rxn)}}{\text{moles (L.R)}} \times \text{coefficient (L.R)}$$

Hess's Law:



$\Delta H = -30\,KJ$

$\Delta H = +30\,KJ$

$\Delta H = -60\,KJ.$

If the summation of step reactions give a main reaction, then the summation of the enthalpies (ΔH) for step rxns gives (ΔH) for the main rxn.

Given the following rxns:

(1) $NaOH + HCl \longrightarrow NaCl + H_2O$ $\quad \Delta H = -56\ KJ$

Strong base. Strong acid

$OH^- + H^+ \longrightarrow H_2O.$

(2) $NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$ $\quad \Delta H = 50\ KJ$

strong base. weak acid

$OH^- + CH_3COOH \quad CH_3COO^- + H_2O.$

- The above rxns are acid-base rxns
- Usually acid-base rxns (neutralization rxns) are exothermic rxns.

Using rxns (1) and Find ΔH_{diss} for the dissociation of CH_3COOH

$CH_3COOH \quad CH_3COO^- + H^+$

rxn (1) $(x -1) \Rightarrow (-56 \times -1)$

rxn (2) $(x 1) \Rightarrow (-50 \times 1)$

$\Delta H_{diss} = +6\ KJ.$ (endothermic)

For the formation rxn:

$$Mg_{(s)} + \frac{1}{2}O_{2\,(g)} \longrightarrow MgO_{(s)} \qquad \Delta H = ?$$

given that:

(1) $Mg_{(s)} + 2H_3O^+ \xrightarrow{x1} Mg^{2+} + 2H_2O + H_{2\,(g)}$ ΔH_1

(2) $MgO_{(s)} + 2H_3O^+ \xrightarrow{x\bar{1}} Mg^{2+} + 3H_2O$ ΔH_2

(3) $H_{2\,(g)} + \frac{1}{2}O_{2\,(g)} \xrightarrow{x1} H_2O_{(\ell)}$ ΔH_3

Applying Hess's Law on equations (1,2,3)

$$\Delta H_f = \Delta H_1 + (-\Delta H_2) + \Delta H_3 \qquad \text{(poor)}$$

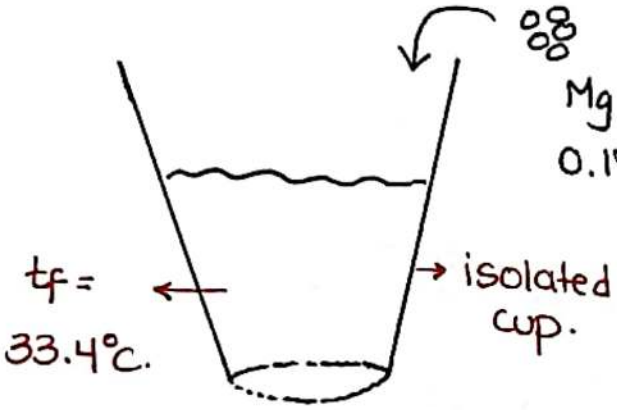
- Note: equations or rxns 1, 2, are the reactions for which their enthalpy is determined by cup-calorimetery.
- The most important is to measure (t_i) before rxn, and (t_f) $\approx t_{max}$ during rxn.

$$Mg_{(s)} + H_3O^+ \longrightarrow Mg^{2+}_{(aq)} + H_2O + H_2$$

$Mg_{(s)}$: 0.17gram

H_3O^+: 25mL 0.1M HCl

$Mg^{2+}_{(aq)}$: $MgCl_2$ solution (new soln)



$t_{max} = 33.4°C$

empty cup = 5.23g.

25mL (0.1M) HCl

$t_i = 24.2°C$

(Surr = cup + new soln.)

$q_{surr} = q_{MgCl_2\ soln.}$

$q_{cup} = 0$.

$$q_{surr} = \text{mass} \times \text{sp.ht} \times \Delta t$$

$$= (25mL \times d_{(\text{new soln})}) \times 4.07 \frac{J}{g.°C} \times (33.4 - 24.2°C)$$

$$= \left(25mL \times 1.02 \frac{g}{mL}\right) \times 4.07 \frac{J}{g.°C} \times 9.2°C$$

$$= +954.8\ J.$$

$$q_{rxn} = -954.8\ J \quad \text{(for 0.17g Mg)}$$

- $\Delta H_{rxn} = \dfrac{q_{rxn}}{\text{moles (Mg)}} \times \text{coefficient (Mg)}$

$$= \frac{-954.8\ J}{(0.17/24.3)} \times 1.$$

$$= -136480\ J$$

$$= -136.48\ KJ.$$

- The rxn is exothermic
- Same for rxn (2)

$$\underset{(0.15g)}{MgO} + \underset{25mL\ (0.1M)}{2H_3O^+} \longrightarrow Mg^{2+} + 3H_2O$$

$$\Delta H_{rxn} = \frac{q_{surr}}{(0.15/40g/mol)} \times 1.$$

q_{surr}: from cup calorimeter.

Experiment (11) Chemical Kinetics.

Rate of reaction = $\frac{\text{Change in conc.}}{\text{change in time.}}$

$= \frac{\Delta \text{ Conc}}{\Delta \text{time}}$

$\rightarrow$ $M/s = mol.L^{-1}.S^{-1}$

(وحدة سرعة التفاعل)

For the reaction :

$A + 2B \longrightarrow 3C.$

rate of consumption of "A" $= -\frac{\Delta[A]}{\Delta t}$

rate of consumption of "B" $= -\frac{\Delta[B]}{\Delta t}$

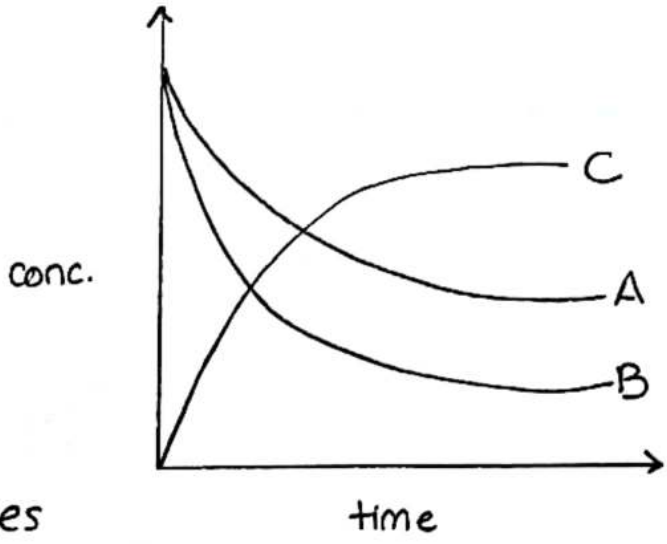
rate of formation of "C" $= \frac{\Delta[C]}{\Delta t}$

Rate of rxn $= -\frac{\Delta[A]}{1\,\Delta t} = -\frac{\Delta[B]}{2\,\Delta t} = \frac{\Delta[C]}{3\,\Delta t}$

$\left(\frac{\text{سرعة استهلاك (A)}}{1}\right)$ $\left(\frac{\text{سرعة استهلاك (B)}}{2}\right)$ $\left(\frac{\text{سرعة انتاج (C)}}{3}\right)$

Scanned with CamScanner

- Rate of rxn is always positive. (السرعة دائماً موجبة)
- Conc. of reactants decrease with time.
- Conc. of products increases with time.



- The maximum rate is at the beginning of the reaction. (أعلى سرعة للتفاعل تكون عند بداية التفاعل)

- Rate Law: For the rxn : $A + B \rightarrow C$
- $\text{Rate} = K [A]^x [B]^y$
- [A], [B] : initial concentrations of reactants.
- x: order of rxn with respect to "A"
- y: order of rxn with respect to "B"
- overall order = $x + y$. (الرتبة الكلية للتفاعل)
- K: rate constant (depends on temperature)
- As temp increase , "K" increase.

(علاقة طردية بين الحرارة وقيمة K)

Scanned with CamScanner

- Unit of "K" = $M^{(1-x)} \cdot time^{-1}$ (x: overall order)

Example : For a third order rxn:

$$K: M^{(1-3)} \cdot S^{-1} \Rightarrow M^{-2} \cdot S^{-1}$$

$$= (mol.L^{-1})^{-2} \cdot S^{-1}$$

$$= mol^{-2} \cdot L^{2} \cdot S^{-1}$$

(وحدة ثابت السرعة) $= L^2 / mol^2 . s$

- Conc. of products does not appear in the rate law expression. (تركيز النواتج لا يظهر في قانون السرعة)

- Activation energy : minimum energy required to start up reaction.

"energy required to break all bonds in reactants". (طاقة التنشيط : الطاقة اللازمة لكسر الروابط وحدوث التفاعل)

- Unit of (Ea) : KJ/mol or J/mol.

Scanned with CamScanner

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{-E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \qquad (T_2 > T_1)$$

T_1: lower temperature in Kelvin.

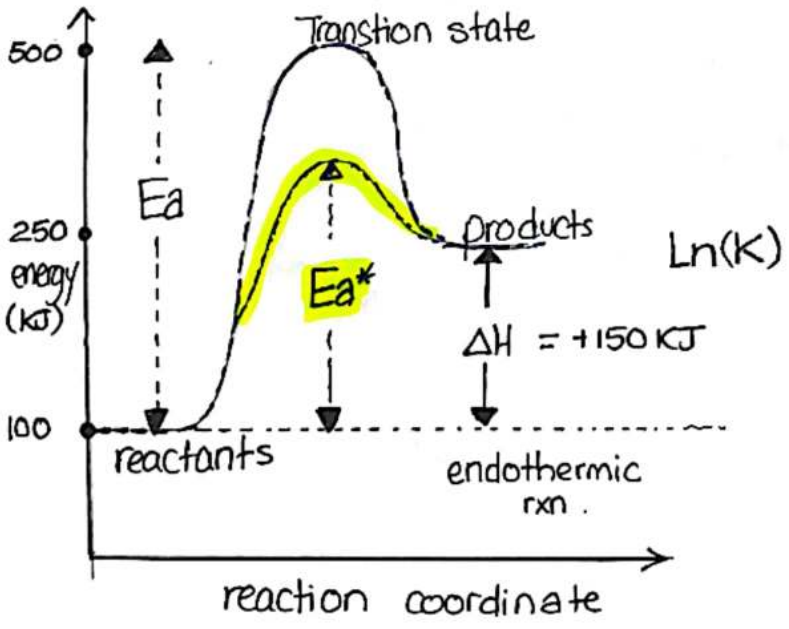
T_2: higher temperature in Kelvin.

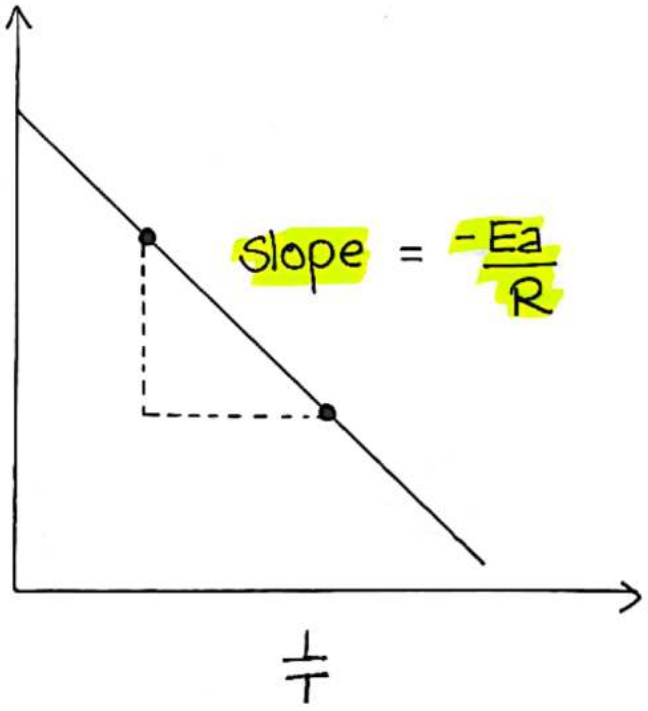
K_1: rate constant at "T_1"

K_2: rate constant at "T_2"

R: 8.314 $\frac{J}{mol.K}$

Ea: activation energy in $\left(\frac{J}{mol}\right)$ ⇒ (+ve)





Ea: activation energy of rxn

Ea*: activation energy in presence of a catalyst.

Scanned with CamScanner

The orders of a reaction are determined experimentally, then the value of K is found. (دُتب التفاعل يتم ايجادها عملياً)

For the redox rxn:

$$H_2O_2 + 2I^- + 2H^+ \longrightarrow I_2 + H_2O. \quad \text{..... rxn (1)}$$

$$\text{Average rate} = \frac{\Delta[I_2]}{\Delta t}$$

But no physical observation is noticed for this rxn.

Another reaction in progress is required.

$$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-} \quad \text{..... rxn (2)}$$

I_2 ↓ consumed by $S_2O_3^{2-}$

$2I^-$ (colorless soln)

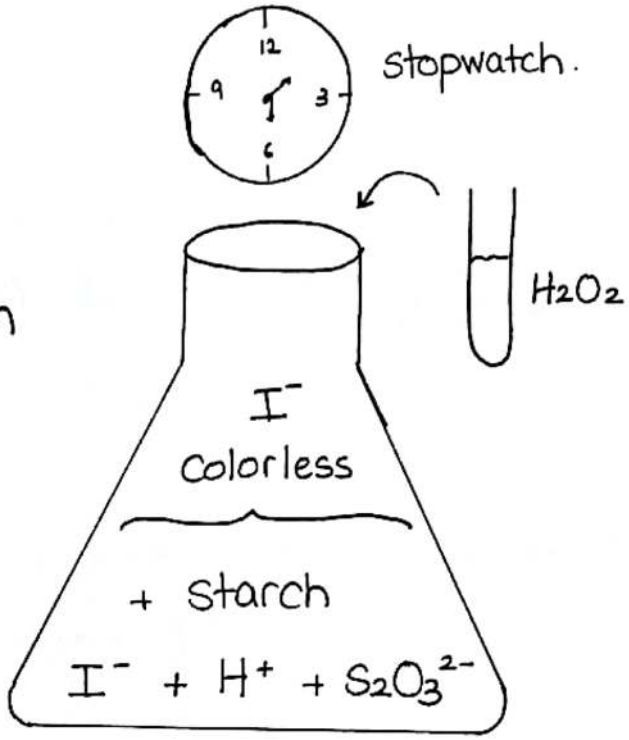
$$\text{rate of rxn (1)} = \frac{\Delta[I_2]}{\Delta t} = \frac{-\Delta[S_2O_3^{2-}]}{2\Delta t}$$

$$= -\frac{\left([S_2O_3^{2-}]_f - [S_2O_3^{2-}]_i\right)}{(t_f - t_i)}$$

Scanned with CamScanner

- From rxn (1):

$H_2O_2 + I^- + H^+ \longrightarrow I_2$.



Then I_2 produced from rxn (1) reacts with $S_2O_3^{2-}$ to produce the colorless (I^-) solution.

(سيبقى اللون شفاف طالما يوجد $S_2O_3^{2-}$ في وعاء التفاعل)

(1) $H_2O_2 + 2I^- + H^+ \longrightarrow I_2$

(2) $I_2 + S_2O_3^{2-} \longrightarrow I^-$

- When all $S_2O_3^{2-}$ is consumed (its final conc. = zero) the I_2 produced from rxn (1) reacts with starch to form the dark blue I_2-starch complex.

- Appearance of the blue color means that all ($S_2O_3^{2-}$) has been consumed.

$$[S_2O_3^{2-}]_f = 0 \text{ M}$$

- ظهور اللون الأزرق يعني أنه تم استهلاك ($S_2O_3^{2-}$) بشكل كامل وفي هذه الحالة يرتبط (I_2) مع النشا ليظهر اللون الأزرق.

Scanned with CamScanner

In this experiment:

$[S_2O_3^{2-}]_i = 0.005M.$

$[S_2O_3^{2-}]_f = 0$ (when blue color appears)

$t_i = 0.$

$t_f = 200s$ (time required for blue color to appear).

$$\text{rate of rxn} = -\frac{\left([S_2O_3^{2-}]_f - [S_2O_3^{2-}]_i\right)}{2(t_f - t_i)}$$

$$= -\frac{(0 - 0.005M)}{2x(200 - 0)\,s}$$

$$= 1.25 \times 10^{-5}\ M/s.$$

So rate of this rxn is calculated as the following:

$$\text{rate} = -\frac{[S_2O_3^{2-}]_i}{2t}$$

t: زمن ظهور اللون الأزرق

Scanned with CamScanner

$$H_2O_2 + 2I^- + 2H^+ \longrightarrow I_2 + H_2O$$

↓ buffer solution. (no effect on rate)

$$\text{Rate} = K[H_2O_2]^x[I^-]^y$$

- $\text{Rate} = \dfrac{[S_2O_3^{2-}]}{2t}$
- x = 1
- y = 1
- K_1 (at 22°C) = 7.24×10^{-2}
- K_2 (at 50°C) = 0.151
- Unit of K: $M^{-1} \cdot S^{-1}$
- Ea = +20797.3 J = + 20.79 KJ

- The total volume in all trials equals (100mL)

Trial	0.1M KI (I^-)	0.03M H_2O_2	0.05M ($S_2O_3^{2-}$)	Starch (mL)	H_2O (mL)	Temp °C	Time (S)	Rate of rxn (M/s)
1.	(5mL) $5 \times 0.1M = M \times 100mL$ 0.005M	(5mL) $5 \times 0.03 = M \times 100$ 0.015M	(10mL) 0.005M	5mL	75	22°C	460	5.43×10^{-6}
2.	(10mL) $10 \times 0.1 = M \times 100$ 0.01M	(5mL) $5 \times 0.03 = M \times 100$ 0.015M	(10mL) 0.005M	5mL	70	22°C	235	1.06×10^{-5}
3.	(5mL) $5 \times 0.1 = M \times 100$ 0.005M	(10mL) $10 \times 0.03 = M \times 100$ 0.03M	(10mL) 0.005M	5mL	70	22°C	242	1.03×10^{-5}
4.	(5mL) 0.005M	(10mL) 0.03M	(10mL) 0.005M	5mL	70	50°C	110	2.27×10^{-5}

Scanned with CamScanner

- Factors affecting rate of reaction.

(1) Temperature. (علاقة طردية)

(2) Concentration of reactants (علاقة طردية)

(3) Nature of reactants (طبيعة المتفاعلات)

(4) Presence of catalyst. (وجود عامل مساعد)

- Remember: Fast reactions have:

High rate, less time, Low Ea, high "K"

- Objectives of this experiment.

(1) Calculate average rate. for a reaction.

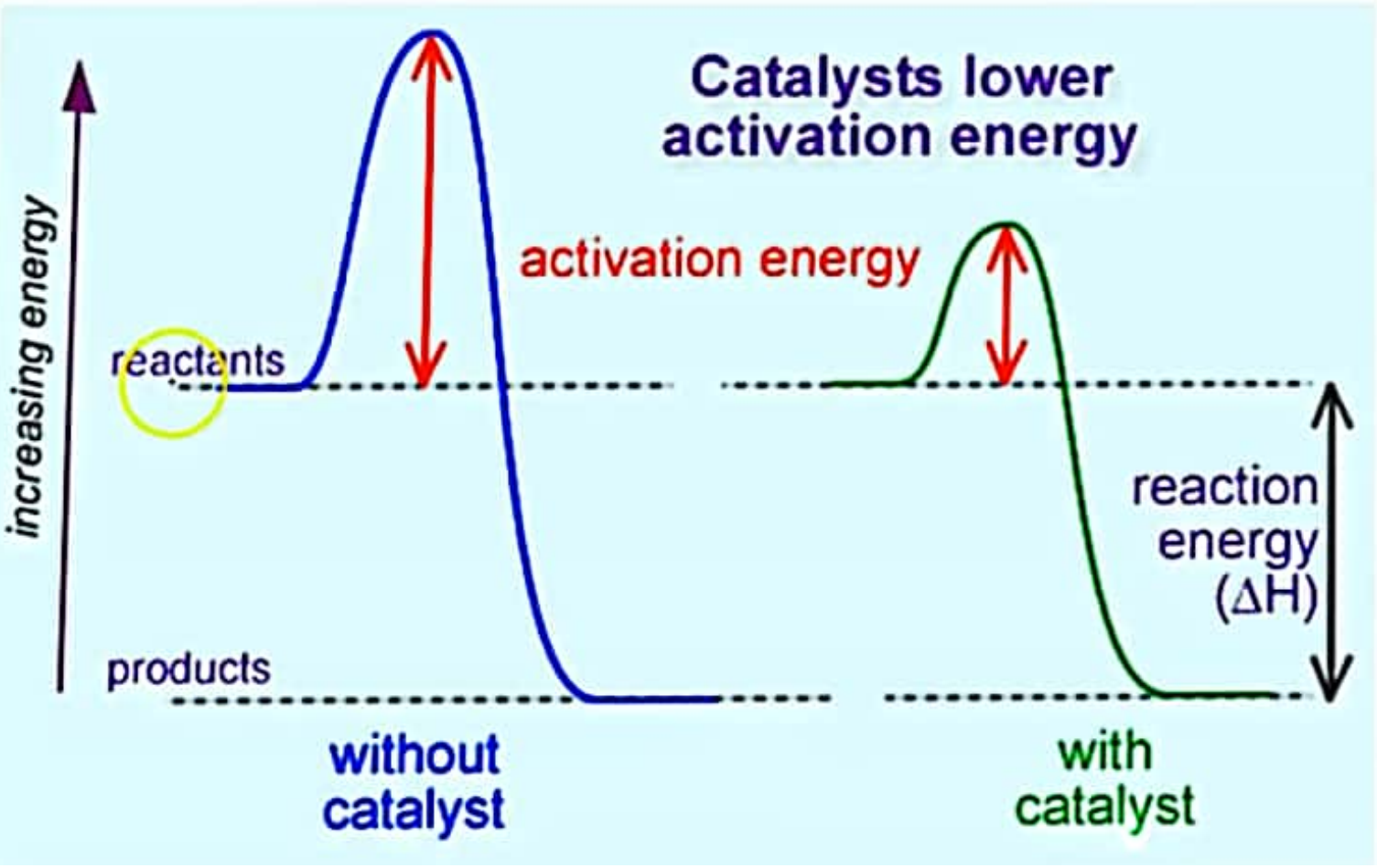
(2) Write the rate law expression.

Find orders, K (K_1, K_2).

(3) Calculate activation energy of a reaction.

Scanned with CamScanner

Factors affected rate reaction:



Scanned with CamScanner